

ActInf Livestream #018.2 ~ The predictive global neuronal workspace: A formal act inf model

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hello and welcome to the active inference live stream this is the active inference lab and we are here in active inference live stream 18.2 on march 30th 2021 welcome everyone to the active inference lab this is a recorded in an archived live stream so please provide us with feedback so that we can improve on our work sorry the video just changed a little bit there we go um all backgrounds and perspectives are welcome here and we'll all be following good video etiquette for live streams today we're here in session 18.2 on march 30th having the follow-up discussion with the authors and other participants on this predictive global neuronal workspace paper and today the goal is really to follow up on the last week's discussion continue discussing and learning about this very cool paper and why the authors did what they did and what are the implications for different areas so today we're just going to go through some introductions and warm-up questions and then we'll be able to walk through the paper and the slides and ask a few questions to the authors and get a few different perspectives so for the introductions we'll just go around and introduce ourselves and then pass it to somebody who hasn't spoken yet i'm daniel i'm a postdoc in california and i'll pass it to alex thanks hi everyone i'm alex watkin i'm a researcher in systems management school in moscow russia also i'm a corner of co-organizer for active inference lab and i pass it to steven hello i'm stephen i'm based in toronto and i do a lot of work with social topographies and landscapes for social drama and i will pass it over to dean hi i'm dean uh i am retired i'm hanging out in the uh loft of my cabin which is not very well lit uh i'll pass it to ryan um yeah um so i'm ryan smith um i'm a investigator at the laureate institute for brain research i'm in tulsa oklahoma um yeah i mainly work on computational modeling of uh empirical behavior and different psychiatric disorders um as well as some theoretical work on emotions and emotional awareness and interception and i guess um i guess i'll send it to chris hi i'm christopher white i'm a phd student at the university of cambridge in england um and i mostly work on active inference or computation using active inference with computational models of cognitive

control but a lot of my previous workers kind of focused on visual awareness so i'll pass it to adam hi uh i'm adam i'm a post-doc at johns hopkins school of medicine um center for uh psychedelic and consciousness research i'm also a research fellow at the kinsey institute and i'm in consciousness free will too that everyone cool does anyone want to raise their hand and give a thought on one of these warm-up questions which are just what are something they're excited or curious about today or something that they liked or remembered about the paper or last week's discussion to sort of jumpstart where we were at um or maybe really quickly at the beginning christopher wanted to clear up something so yeah and then we'll kind of go into questions so maybe everyone could write down a question and then christopher go for it yeah sure so i said two things that like just false um last time i just made mistakes i kind of wanted to correct them so the first one i was in specifically referenced someone asked a question about the thalamus and its involvement in consciousness and i kind of referenced a whole body of work from matthew larkham's group and i think i said the medial dorsal nucleus and it's not it's central lateral nucleus and thalamus um i think like 99 people wouldn't care but anyone who knows anything about the thalamus would have been like pained by that um and the second thing was i was talking about king and hans like uh bayesian model of conscious access i think i described the model as univariate gaussian that's also not correct it's a multivariate um like gaussian although you can extend the same model um to a univariate case anyway doesn't matter but i just wanted to kind of correct those two things thanks for the correction so if everyone could raise their hand and first um we'll go with adam and then stephen um i was uh curious about uh what chris just just said so like medial door so that would be what thalamus for interior singular in the frontal lobes kind of like the gateway to um like amygdala at all would that be right and then like so the central ladder like who's that looping with i forget i think central lateral would be looping i was specifically thinking that like a loop between layer five pyramidal neurons and um central lateral thalamus so if i could ask a question there just why are we identifying brain regions or what would it mean to be identifying brain regions one way or the other in humans or some other species um so this particular body of work not to go too far on a tangent but there's this really impressive body of work that basically has identified both level of consciousness and content of consciousness with burst mode firing in layer five of cortex i think most of this at most of this evidence is from a mouse model though so interpret that as you will yeah adam or you can use the actual jitsi bracelet but yeah go on um i mean for me um it would partially be what systems might contribute to wakefulness or different aspects of consciousness depending how we want to

carve the joints on it in different respects so um let's say uh stimulating the medial dorsal was like more important well then that might implicate like the specific relays um its ability to form phlegma cortical loops or flammable striatal core like the whole uh what it helps to bind together functionally and other functions uh it would tell you what processes might be involved so um maybe action or salience that might be important or maybe it's like frontal lobes as hub but like which part of the frontal lobes is it the parts that are like a general topologically central hub or does it look kind of like a motor bits or does it look like something else so i'd be trying to look for like fish around for clues about what's important in what ways but it might mean but it's easy to tell stories okay cool so stephen and then anyone else who has a question so yeah i'm just uh really excited about the when i when i listened uh last week you talked about temperature being used in the models and like you had different places where you used different amounts of temperature in i think it's in the soft map function to to show um different types of model scenarios so i'll just be curious i'm just excited about maybe um just the general way that you use your thinking to build the models to inform how you start to look at consciousness and the way the brain works as much as the way that you find the results from the models so that i'd be interested in that those sort of two sides so so i mean one one thing i just want to make sure this is clear is that you know like computational neuroscience borrows or has been historically ended up borrowing a lot from um terminology and physics and engineering and things like that um you know so there's some of these sorts of terms that can be a little um a little funny right so so like temperature like a temperature parameter or typically an inverse temperature parameter um there's something that's borrowed from physics that just kind of has to do within the physics set in the physics sense kind of how how much uh you know with how much energy particles are kind of bouncing around against each other um and um you know in a very oversimplified way of putting it um but but so it just means essentially there's more kind of randomness right in the in the system so in the context of um of this um you're just applying that same equation to describe essentially the amount of noise in decision making so uh a higher temperature in this case just is just a parameter in an equation that controls how deterministic versus stochastic decision making processes are so i just want to make sure that it has i just want to make sure it's clear to anyone listening that it has nothing to do with actual like physical temperature like being hot or cold it's just a parameter that just that kind of modulates how random decision making is um so i just wanted to make that make that clear so in this case it's just it's just controlling so so for example if if the temperature parameter is really low or the inverse temperature has a really

high value then that just means that say even if the model was only like 51 versus 49 confident in one thing over another it would always choose the action associated with the 51 um whereas if the temperature value temperature parameter value is high or is low um or is high or the inverse temperature parameter is low then um that means that the like for instance uh that with 51 of the time it would choose the um the action associated with 51 percent proper with 0.51 probability you would choose that one less often than it would if the probability would say 0.7 um you know and less probability than that than if it was if the probability was 0.9 so essentially just um having a a low inverse temperature value just allows the probability of choice um the actual choices uh that's our frequency of choice um to kind of scale in some sense with the actual probabilistic beliefs over the different um states in the model thank you ryan we'll have christopher then adam and anyone else who raises their hand so just to kind of contextualize that a little bit um so a soft max function is exactly the same function as a boltzmann distribution from statistical physics so often if you have a you'll just use the same terminology because it has and in the physics case it has like a literal physical interpretation in this case it doesn't um so that's kind of why there's that overlapping technology um in terms of deciding why or on those parameters or actually building them in it was really just kind of came from thinking about what perce what attention does so what we're trying to modulate there was a tension and signal strength and both of those things in some way modulate the strength of feed forward input so when you think about that you then kind of think okay what part of the model corresponds to that well it's the likelihood matrix that maps but that decides the precision of the mapping between external observations and first level hidden states and so both the idea is is that both attention and kind of external signal strength where or it might be something like contrast whatever um will jointly determine the precision of that mapping um and then there's lots of different ways where you can kind of cash that out if you actually want to get into kind of the neurobiology of it like the normalization model of attention or bias competition or something like that but kind of extracting away from that detail basically it just modulates the a matrix the computational level of analysis so that was kind of the thinking that went into it very interesting adam then stephen than anyone else okay adam can you be ready to speak but go for it yeah sorry about that oh um it's kind of off the current topic but from before so some quick googling uh turned out the central lateral um seems to be looping largely with parietal and temporal association areas so that's potentially interesting in ways that indirectly could like speak to debates on the physical substrates of consciousness like the the brouhaha between iit and gnwt said more like the front of the brain or the back of the brain

and in which senses do they mean but it's interesting i actually i'm having trouble finding an image of like a diagram of the thalamus actually shows the centro uh lateral nucleus it's all just like i see central medial but not central lateral it seems like they might be kind of like tucking it in with the pulvinar whatever that means stephen yeah just say thanks for that that feedback that was really useful i think also i because i've got a background in chemistry i keep thinking about the gibbs free energy where it talks about um temperature times change in entropy in that term so it's like um that that the temperature is multiplied into the amount of entropy change there is to make it more significant so that would kind of mirror why it's a bit like you're putting in noise or entropy so uh that that's quite useful um so yeah thanks yeah welcome blue you you um you or anyone else can raise their hand but one chemistry metaphor and again ryan as you brought up it's not always like a literal uh meaning but thermodynamic versus the kinetic alternative for a reaction so like the thermodynamic outcome for the candle is to burn because it has all this pent-up energy but it doesn't spontaneously combust because the kinetic barrier the activation barrier is too high and then when there's like an enzyme or a spark of energy then at in the right situation the product can go all the way down the big delta g to the kind of ultimate thermodynamic least energy versus the more kinetically accessible and so that's kind of like information foraging there's sort of the perfect citation you would find if you had the right search and it was that low temperature or something analogous but then if it's too um kinetically constrained then it's the attention is drawn away i'm not sure if that will map directly in this whole active inference mode but it's just kind of another chemical informational distinction well i mean it depends i mean it depends how you're using it like in this in this case there's not really much of a connection because we're just essentially we're just using a temperature parameter and a softmax function to control the precision of a mapping well in a few different places for attention and for stimulus strength we're just using it to change the precision of the mapping between states and observations at the first level um so it just it just controls essentially how how strong a sensory signal ends up updating beliefs um posterior states at the first level um and then modulated by a second temperature parameter that does the same thing but that accounts for attention so these sorts of things jointly um control um essentially how much evidence sensory input provides for posteriors over states at the at the first level um you just we're using it for that kind of thing and then we also use it for controlling deterministicness or the uh at the second level of the model in terms of in terms of uh action selection um so in in all those cases i i don't personally see a direct relationship between that and the kind of thermodynamic um thermodynamic aspect of temperature of

getting over um you know energy barriers and things like that um but i mean it could be i could be uh i could be just not thinking through it all the way stephen then anyone else yeah i think that makes a lot of sense what you're saying and i suppose the nice thing with the modeling is that you've got this potential to i know you're using it like you say is this way to bring in noise or soft max but it gives you that way to increase and decrease it which in real life when you do empirical work it's hard to to do that so it's it's one of the strengths of the modeling that you can change that parameter and see it seems to be one way that you really get insights into the dynamics of the system um yeah i mean in terms of just being able to modulate modulate the uh or mimic the influences of attention on modulating the background perception of the signal yes stephen or uh dean and then stephen yeah i just have a quick question here and maybe this is far too oversimplified but when you guys were doing this work i i started also looking at the uh um the effect paper that i think the two the two together gave me a better sense of some of the work that i was doing around helping people walk into these novel situations pre-reflectively and there is a tendency to want to commit to want to pay attention to want to give over whatever you have in terms of processing power to figuring out what's going on but we had to spend quite a bit of time on helping people get the idea that there is this channel there's this kind of sweet spot and when you were talking about degrees it brought it up for me again the idea how to help a person know when they're over committing when they're over correcting and when they're over compensating so when they're trying to do this estimation how do they know that they're not seeing the moon walking bear because they're so committed to counting up the number of basketball passes i wonder if you guys had spent a little bit of time looking at the other end of the of the channel when people can actually readjust back in and i know that's that's probably where the degrees part comes in but i was just wondering about that so i mean are you imagining are you imagining in some kind of like clinical scenario where a person yeah like biased attention away from something that could be say aversive if they were to focus on it yeah exactly um yeah so i mean i mean there's there's probably a mechanistic relation in a sense just in the broad sense of you know how we think uh attention is implemented right so so for example if you if you think that you know a person um is avoiding paying attention to something because it would be aversive right then then you can end up with this kind of implicit reinforcement of attentional policy selection which we don't directly have in in this model right the agent is choosing what the precision is associated with the tension in this model but but that's something that can be added and something that we have actually added in some more recent work that we're doing right now um but the idea

the idea there or one way to think about that would be that when uh whenever a person essentially starts to pay attention to the thing that would be aversive then they start to feel it starts to feel aversive right so then if they pay attention away again then the the choice of paying attention away actually ends up being negatively reinforcing because the bad feeling goes away right so then what that ends up doing at least according to this kind of reinforcement learning account is that that ends up assigning higher value to the policy of attending a way right so over time the probability of attending a way just ends up getting higher and higher because it makes you feel better every time you do um so then this sort of avoidant attention can become really kind of habitized um and and uh so you know that's one way of thinking about it and so you know and like in like you know in in the clinical setting right you you want to help people potentially you know like start to you know gradually you know expose themselves to to you know pay attention to this thing that's aversive um in hopes that if they're able to kind of do that in a sustained way then over the long run the aversiveness will go down as they kind of learned a better deal with the thing that they've been avoiding intentionally um you know so so that's um that's um yeah fairly tangential to right the kind of thing we're trying to do here but um you know but there's a there's a connection in this very kind of broad sense in that if you can control attention and choices of what to pay attention to can be reinforced then that's one way you can kind of end up with this this kind of thing where person might not even know right that they've developed this strong habit you know not to pay attention to something it just becomes very automated um you know the same way the same way any other behavior can be habitized via repeated reinforcement um but here just it would be a kind of a cognitive action as opposed to a a a more overt behavioral action okay very interesting thanks for that question response adam and then anyone else who raises their hand i guess to come out of left field um and what you're just describing it seems like that could have um some relation to maybe grounding like psychodynamic-like concepts in terms of if these patterns of attentional approach and avoidance can move from your c matrix to your e matrix you can end up finding yourself in these sort of like garden paths of attending and coming to uh inferences for reasons that you might not even understand just on the way your policy selections of attention have been shaped for conditions that are might be too uh numerous and varied for you to track and so yeah yeah i mean there's i mean we've written you know we've written a few papers on this right like or i have with some colleagues i mean again you know my focus um you know is largely in you know in terms of and psychiatric conditions right and i wouldn't um i would say it's a way of of recasting the sorts of behaviors that psychodynamic

approaches are trying to capture but to do it in a way that doesn't um actually posit the existence of these sorts of repressive right like you know like unconscious like there's no unconscious agent right there's no id right that's being positive that's sitting there keeping these things out of awareness but somehow they're still present unconsciously right there's nothing like that right it's just a very simple story of having reinforced patterns of attention that um you know so it it's a way of capturing the the phenomena that that i think psychodynamic approaches are often trying to capture but it doesn't posit a lot of the kind of like ontological um a lot of the entities that are positive in psychodynamic um so yeah it's like yeah i'm not saying like you're freudian but like yeah you're actually finding like the baby in the bath water rescue eyes rescuing it cleaning it up contextualizing it like why did people have these intuitions to begin with like what was the grains of truth um i'm wondering if through this this could lead to um like this kind of uh patterns of attending some of which you're not tracking this could be like an account of what would lead to potentially uh mismatch or discrepancies between things like stated and revealed preferences or a thing like um your actual affective disposition with respect to thing and then what you think is going on and then those potentially go across purposes in ways that could be different difficult to explicitly model and potentially have problematic sequela or not but i'm wondering if like this could lead to like a mismatch or across purposes of uh attracting states at different levels um i mean i probably need more concrete examples to try to figure that out i mean that's probably something i'd have to think about more i i guess one quickly would be like um implicit affect tests or something like that like you say like um uh i'm not racist but then it turns out no no you are and like if you actually look at the reaction or something like that or i am really happy in this relationship but then like i measure your heart rate variability and its crap or things like that um just this your actual affective state might end up being different than what you think it isn't there could be like a lack of self-awareness like a difficulty coming to coherent modeling of yourself because of these shaped paths of attention like you might like miss the mark about you because it was like difficult to see something like that i mean i mean i i would i would think that you know i mean perhaps there are sort of some instances of the sort of thing you're talking about to be captured by the um the kind of mechanisms that you know we're talking about here or the kind of tangential extension of them that we're talking about here um but i'm not confident that all of them could i imagine there's a number of mechanisms that you might feel to describe some of those things like um you know implicit yeah like i mean if you're talking about like implicit associations on like like an iat right or something or something like that that's probably my

intuition is that's probably different from um yeah it's more kind of like a like like uh like us right which i know chris has actually been working on models of street tasks um but um right which is which is probably pretty different from um um some of the other examples you were given but but yeah i mean like i said i'd have to taking a particular example i probably have to think it through a little more um to see think about what sorts of what kind of generative model would be able to reproduce what kind of behavior you're talking about um or or you know most likely there would be multiple generative models that could reproduce it and then you'd have to do some kind of experiment right to do model comparison and see which one actually best accounts for it um but again i wouldn't feel really confident um you know saying what that might look like um you know just just in the very kind of abstract general you know description we're talking about right now nice nice thank you for the response stephen yeah i'll just bring in this sort of this point with this model that's now sort of maybe getting into the model in part is i think that's interesting with the awareness base there's a lot of work with awareness-based systems change and looking at awareness and i i like the um i'd be interested in your thoughts on how we can move beyond being trapped in a psychology of the individual and you know pathology of the individual and it moves into the inter-subjective multi-scale so like you talk about tasks but you it also could be used in task scapes you know organizational contexts or groups of people so i just think that there's um because the the it's not nailed down to like a psychological model of the person's thoughts and feelings but more a general model of the attentional structure it has like a you know ability to move beyond between scales and context so um i just think i'll put that out there as something that seems quite interesting um i mean there's more i mean um so i think the utility of models like these is to identify a very specific phenomenon and that namely having conscious access to some content and not others um so i have conscious access to what's in front of me my visual field at the moment if i direct my tension around my body i can bring up all sorts of like i can suddenly become aware of different kind of some other sensory bits of information and all of these have very specific kind neural signatures as it were what we're trying to do is really propose a model at a computational level of analysis and algorithmic level of analysis that explains these and i outside of that i'm not sure to what extent actually talking about awareness into subjective awareness is at all the same thing as talking about awareness as someone who works in like neuroscience of consciousness for example they might be but i don't see any evidence that they are pretty interesting because it's in the title of the paper so many people might think of it differently as i mean it's just an interesting question right sorry what's in the title of the paper

um just the idea that the uh visual consciousness is involved well but i mean but again i mean this is a model of a particular visual past right i mean so it's meant to account for things that are seen in the visual consciousness literature um you know there is the structure is abstract enough that you know we would and we say this in the discussion that it ought to apply to cases of having access to one bit of information versus another um in other domains right like an auditory domain or you know something that i've been working on you know with a bunch of colleagues at um you know at libra where i am um you know it's trying to see if the same sort of thing it looks like it's present in the interceptive domain for example um you know so so i i think it's right to say that you know we would we would hope anyway that it's abstract enough that it can account for any any piece of information that you either gain access to or not you know when it's being represented so much so right so whether it's you know right now as i'm talking to you i'm not necessarily aware of what my heart's doing but now i choose to pay attention to it and all of a sudden i can kind of feel that it's beating at a certain rate i believe i believe i believe that i feel i wasn't conscious of that before despite the fact that my brain was unconsciously tracking you know what my heart was doing that whole time right so it could apply to being aware or not aware of any stimulus so i mean to the so only to the extent when we say something about intersubjective awareness only to the extent that that means something about a piece of information that you know before i was attending to it i didn't know you know was there but now that i'm paying attention to it that i do so for example like if i happen to notice that a person's smiling or something like that um even though they were smiling at me the whole time you know maybe my brain had picked up on the fact that they were smiling but um i wasn't aware of it until i focused my attention on it or something like that right or even maybe more abstract than that you know when my brain sort of had unconsciously detected the smile there might even have been some further inference that that means something about them being happy right so so there might even be a case where um i either do or do not gain access to this unconscious inference or this posterior belief that the person is more likely happy than unhappy or something like that right but uh but i mean these are sorts of cases they would have to be tested in their own in their own right my broad point is just that it's an abstract structure where all it is meant to account for is when your brain is representing something and you either become aware of it or not i mean i mean if i could just kind of build on that slightly so i think one it's very tempting when we've got computational models that are real when we've got lots of equations sitting in front of us to kind of think that i think it's easy to get an illusion of rigor where there is none um the thing that makes these

models meaningful is that we can we're tracking some real world phenomenon um in terms of the abstract model structure it may be the case that say you have group dynamics where there's some group that some aspect of that maybe the individual behavior of individuals within a group evolves at a really quick time scale and then the decision that group decision making evolves as a slower time scale you may be able you you could in principle fit a two-level partially observable mark of decision process to this as a model of that behavior but that would be very very diff i think even in that case even when we're fitting the same model i don't think that those are at all the same phenomenon although there may be similarities across them um i would be extremely resistant to calling one visual awareness or but a group has visual awareness interesting and of course a parallel debate in the ant world is there a colony awareness or a single six-legged ant awareness or how should we think about distributed systems and aren't we all kind of just distributed systems all the way down so adam and then stephen um so this would be related to i guess uh different types of uh interest subjective or yeah interest objective awareness that you might believe is a model for um i recently um uploaded a preprint after some conversations with people from the brain institute at chapman university basically trying to do the beginnings of like an active inference account of la bay phenomena and readiness potentials that are a little bit less deflationary with respect to uh volition and uh conscious causation and so like in like essence i was thinking of like um the actual uh moving your hand being like an accumulation of model evidence model evidence with a proprioceptive pose and it might be some sort of like conjunction of like um also like and hum like whether or not your like will or feeling of urge was part of it you could think of that as like um inference over your affective state and that if these things are in what kind of alignments could help to explain whether you're like aware of readiness potential activity or whether this awareness is actually feeding back and contributing to it and so i feel like your model could potentially like provide like a very precise contextualization of this really subtle what's going on uh then i would like to collaborate so so i mean just just just again brad just to make it really maybe that could be the case i have no idea but it could only potentially be the case if what you're talking about right like what you're you know motivated to do or what you're you know in the process of about you know being about to do or you know whatever aspect of um you know action or action tendency or you know anything like that um the only way this could apply is if you think that that information about what your you know what you're motivated to do or or something like that you know whatever it is and only if it's the case you think that is a represented piece of information um that you can attend to or not um

and that then you could sort of um gate into the higher kind of people deeper temporal level of the model that would inform um verbal reporting and sort of other goal directed uses um so it'd only be the case i guess my point is this is if the the representation associated with action that you're talking about is in some sense formally equivalent to a representation of something visual um that that you can attend to that makes sense so it's um so the preprint is like it's like a three i think like a three page more of a promissory note but if i could uh like pick your brain later about like if you think there's like a way we could actually get some like empirical traction and like nailing down the specifics along the lines you're saying so well if you haven't i mean if you have a task and that task involves people paying attention to the aspect of action that you're talking about um then in principle you could have people you could have people you know in some sense cue them or prime them or something like that that that engages that sort of action bias or action motivation um and again you can they can either attend to it or not or report on it or not in that case yes but otherwise probably not it's a little tricky i think in terms of like the action that would be attended to would be of a couple varieties but none of them through them would be um directly observable so it's like you could say like it's the action of the neural activity in terms of like the ramping activity like do they have access to that but the i was thinking it could also be like um the action of like mental acts and like rehearsing like imagining a kind of sophisticated effect of infant sense like am i gonna move my hand or not what would that be like and then that being contributing to that signal um but and where like actual awareness of this like fictive meant this fictive action could be part of what would contribute to uh whether or not you feel ownership of it like it's it's interesting like some people don't show readiness potentials they just act other people do um like schizophrenic patients they tend not to show readiness potentials before they spontaneously raise their hand but it would be hard to get this um the action is internal and so i don't know what you should do with that yeah it just seems it just seems like you have to have some kind of behavioral readout that can be fit um to unlock to to a model of this to be able to say that this explains the ability to consciously report on it or not right so i guess could you use like like right before leading to like the action there tends to be like it starts out more bilateral and then the ramp activation tends to go lateral and then the hand moves could you use like that sort of signal as the readout like the ensemble activity so in principle you can use you know you can fit models to neurologic neural and you can fit models to anything that generative model generates but but i'm not necessarily all that confident that you'd be able to justify the claim that because the because the model in one generates you know like a readiness potential

signal that that has anything to do with having conscious access to something any of the other problem like if you're using like the ensemble activity it's still the interpretation like awareness of what like what what are we even talking about with the awareness the whole point is to try to like contextualize what those signals mean and so yeah yeah i mean i mean in the majority of cases you know either directly or indirectly you know like verbal report right is still kind of gold standard for whether someone has access to something right even in no report paradigms you're still later right asking them to report what they were experiencing at the time um and so it's i mean in the yeah the issue is that in some sense all kind of supportable or accepted sort of evidence for being consciously aware of something um is always ultimately and chris you can correct me if he thinks is wrong but but i mean i think that it always ultimately comes down to verbal report or some reliable correlate of verbal report whether or not yeah i agree with that so i think i'm not an expert no report paradigms but to my to my knowledge all of them the reason we trust no report paradigms is because our measure whether that is a particular mode of eye tracking or it is just like incidental memory afterwards um all of those things correlate really really well with report and that's the thing that kind of like makes us trust them as measures of conscious access independent of report um yeah i think it's always reports are foundational unless we have really really good evidence otherwise i think we have to take them kind of at face value cool we'll return to the stack so blue and then stephen and then anyone else who raises their hand so adam always brings me into this like metaphysical like thought space whenever he talks um and i just was wondering uh you know in terms of this like visual model there's you hear the expression that people see what they want to see right and so have you thought about maybe testing like you know like there's the the idea of paying attention to something and so we thought about testing like whether you know people can deceive themselves in this visual process right like some self-deception or even like the you hear of like the power of manifestation like can people bring into their awareness something because they simply want it to be there i mean have you thought about this or testing the model in this way yeah i mean if you go to a figure um scroll down is it which one so we have it we have us actually i'm not sure uh so that's the erps yeah that the tax of the extended taxonomy so we start with this kind of four-way taxonomy between which tahan proposed um along with some other colleagues of the kind of factors underlying whether you'll have conscious access to a stimulus and you kind of broke it down into whether attention was weak or absent present or absent or whether kind of just how strong the signal strength was with the streak um weak or strong and then we extended this by adding in prior expectations so the

extent to which you expect a stimulus to appear um and there is now like a one of the big motivations this work is that there is an enormous body of work now showing that expectation does play a fairly fundamental role i think in determining the contents of visual awareness um so as for whether you can voluntarily bring something into awareness i'm not the answer is i think that's more to do with attention i suspect but say for example if you are looking at a bi-stable image you can attend to one part of that image and then have the if it's a i think it's called a neca cube you can have the cube flip versus if you attend to another part it'll flip back and you can do kind of similar deliberate moves of attention with a lot of bi-stable images um so also i think there's some really cool there is some interesting kind of back to the expectation thing expecting seeing what you expect to see a lot of that depends upon how strong your bottom up evidence is if you are much more likely so there are paradigms that kind of distract people but basically condition them into expecting a stimulus to appear um and you can get kind of accidental hallucinate or you can basically induce people's blood what is essentially a hallucination in those cases or mistakenly report something that we know as a fact the experimental as experimental as they weren't there i think there are two examples that what i have in mind at the moment is a really clever set of studies by john aru but i know ryan and maybe you can talk about this there's also a lot of work on conditioned hallucinations from a group at yale who i know you collaborate with yeah so yeah i know that was yeah something i was actually going to bring up is and this is this is not yeah because i mean i agree with chris and in most cases i think it's going to have to do with some kind of like motivated selective attention um that kind of like biases say different possible uh interpretations that you have posteriors over states for the posterior to the first level um but um but yeah definitely you can have these expectation effects although i don't necessarily think of those as being um voluntary right i'm not sure that um like like selective attention is definitely a kind of controllable cognitive action whereas shifting uh prior expectations i'm less confident that can be cast that way um but but um priors definitely have a an influence right on what your posteriors of our states are given some stimulus especially like chris said when the stimulus strength is fairly low or noisy um you know so so some of the work that um yeah al powers and his group at yale and then that i've been we've been collaborating to build active inference models for this stuff um is uh yeah these sorts of condition hallucination paradigms in people's psychosis so the way these paradigms work is that you um you start out always showing people a light um and is always coincident with a tone that appears in some white noise um and so on the first several trials there's a fairly strong tone that plays in the white noise every

time they hear a light or every time i see a light and it's thresholded so that they would perceive that tone uh 75 of the time um and then and then after several of those trials um there starts to be cases where um the the tone is still present but it's weaker so they only would detect at 50 or 25 of the time and then there are trials where the light comes on but no tone is played and there's just white noise um and what um what they have found is that um people with psychosis um have a higher probability of reporting hearing tones when the light shows up even when there was no tone um so let's just say kind of expect so they build up this expectation to hear a tone of every single light and somehow those prior expectation that those prior expectations that the tone will be there um have a stronger influence over whether they hallucinate the tone or not that influence is stronger in people that have hallucinations than in people that don't have hallucinations which has kind of led to this idea that what's what's going you know part of the computational issue and people with psychosis that are more likely to have hallucinations is that um prior expectations are in some way really precise and therefore going to dominate um perception um and that could be that could be either because the prior expectations themselves are really precise or because beliefs about the actual sensory signal are those are believed to be uh very imprecise um but in either case the relative precisions um are supposed to be thought to kind of favor prior expectations and psychosis to a greater degree than people without psychosis thank you can i just follow up really quick yep yeah so it's interesting that you brought the auditory um thing into this but the auditory hallucinations and i wonder if it's a visual cortex thing or if it's involved more with the auditory cortex because even when chris was talking you know you think about like people have seen like when you have two things played simultaneously auditory tracks that like it's like listen for the phone number and then you hear the phone number but then if you're not listening for the phone number you're hearing something else so it's interesting that the auditory system like the the selective attention in the auditory track really is is much more apparent like where you see the blue dress versus the white dress or the gold dress or whatever color it was like the two dresses is not so selective it's like look for the blue dress or look for the white dress if you set up that expectation in the visual system it's not as um prevalent i think yeah i mean i will agree that it's definitely harder to switch the uh interpretation from the blue to the yellow when i've tried that i can i can get myself to do it but it's way more effort for you right um and uh yeah i mean i don't know chris if you have any ideas about why exactly that would that would be i mean i know cases of like you know playing binocular rivalry and things like that in those cases i think there are cases where i think selective attention can play a bigger role like the

neckercube for example um that's maybe a little more analogous to kind of like uh multi uh channel auditory input i think it just depends on how you set the task up i'm not sure if the comparison is really a fair one i suppose i think you can very i think you definitely can set up a lot of expectations play a huge role in the visual system i think is one thing to say um whether you have voluntary control whether those things come under the purview of like voluntary attention is another thing um but i still think there are lots of cases depending on how you set up the problems i think binocular rivalry is a good one or even something like actually literally like where you move your eyes i know that sounds trivial but whether your eyes are moving around a page or not kind of determine determines uh some different types of visual illusions so if you think about um maybe illusion isn't even the right word but what's the where sorry we're working on the project about this right now what's it called again tom's power is a paper on this run do you remember what's trucks were fading yeah yeah yeah if you google troxler fading i think that's a really powerful example of where just kind of selective attention has a really major role on whether you perceive something as yep very interesting it's almost like there's ambiguous and non-ambiguous stimuli in different domains and maybe even interpersonal variation one person being really good at picking up on something subtle in one domain or so we'll do steven then dean then adam i think this ties in actually to what we was mentioning earlier um about you know when you scale things up and like and i agree with what you're saying when you scale things up it may be useful to think about how someone facilitates the attention of people in a group and how you take them and how they you think about taking their attention it doesn't mean that whatever they consciously have inter-subject together is now read by the model you know so it's usefulness may be in how people's attention is taken and i think it ties to this hypnosis thing as well because i think your your model speaks to top down and bottom up happen at different level there are different scales because often we think top down we think executive function and like executive functions telling me what but like you say you can have a subconscious level of trying to think what something is even if it's in the subconscious you can and hypnosis is doing that hypnosis is playing the game with someone um so they they um or with these optical illusions you say to someone this you're looking at this cube from above or below and it flips but it's not that you you're being told it's this type of cube and your executive function's pushing it it's it's still top down but it's it's happening at a um a more subconscious what we might call a subconscious level and i think that's quite interesting so if i could just speak to that i think there are a couple of things going on there so i would talk about executive function well i

mean i think it one thing that this is a bit of a try point but top down is a very ambiguous term um i think what where so you can mean kind of that can mean anything from basically like recurrent loops from higher chords from cortices that are higher up the cortical hierarchy um to something like top-down attempt like a voluntary control of attention via executive like fronto-parietal networks or something like that um so those are just different different phenomena um i think in our model control of would presumably be at if if we were to build a model of this would be at this you you would put the dorsal attention network at the second slower time scale of the model i think maybe ryan you can we can speak this as well i haven't really thought about this um i would have been a little more uh like executive control network um sort of thing i guess would have been a little bit more what i would have um thought about just kind of off the top but um yeah you're probably right i mean visual i mean visual attention right i mean obviously right the dorsal attention network is you know kind of defined or originally yeah like linked specifically to uh uh yeah like visual attention and things like that or or kind of quote unquote top-down visual attention but um in terms of the kind of directing right visual attention in a goal-directed way um then i think i think that would be um more executive control network for that i mean on the hallucination point i i did my undergrad at macquarie university and they have a whole hallucination group there within the cognitive science department and so i i just feel like acutely aware of the fact that i don't know anything about it um and it's also an extremely complicated phenomenon you can use hallucination to like as a model of various delusions like you can uh induce is it called like mirror basic like mirror agnosia where you don't no longer recognize yourself mirror self-agnosia or something where you no longer recognize yourself in a mirror like it's a really powerful tool but i have absolutely no idea how it works and so i would be very kind of i think it's very tempting as kind of modelers and theorists to kind of go into various experimentalist domains like hey here's how my overarching framework explains your thing um and i i kind of want to resist that um i know a little bit about visual consciousness and how expectation all that stuff works there but hallucination is such a hairy topic i just don't know yeah i mean i would also i would also say i mean i mean a few different things have been brought up here right but i mean in terms of uh it's not it's not clear for example but the um kind of auditory hallucination um story that i was talking about earlier with um um in terms of having relatively you know more precise prior expectations um you know creating a stronger tendency for auditory hallucinations and it's not it's not clear at all that that kind of story mechanistically is generalizes to other so i mean actually in the individual domain there's this whole other body

of where it looks like people that have delusions um again as part of the sort of as part of right actually show the opposite weighting in the visual system right so where they actually um assign too much precision to low-level bottom-up sensors bottom-up visual signals you know so they end up as these interesting cases where they end up um if you look at eye tracking they end up doing um people with uh delusions um end up actually or psychosis schizophrenia um end up doing uh better at certain tasks than healthy people um but worse at others so like one example of this is um if you have people just kind of try to follow just visual tracking fall a little kind of moving target um then if that target switches directions really quickly um then [Music] the people with psychosis end up following uh following it better um than healthy people um because healthy people kind of continue to follow it for a little bit um after the turn and the kind of predicted direction um right so prior expectations are kind of playing a stronger role um so they actually do a little better whereas if instead you have something where the um where the target temporarily kind of moves behind some kind of barrier um then healthy people do better whereas people with schizophrenia kind of you know lose it because prior don't do enough um to keep them kind of following what the trajectory would have been right as it goes behind the visual occlusion and there's there's lots of other cases um for instance in like the proprioceptive domain where people talk about the same kind of thing where people with um schizophrenia actually do um better for things like force matching illusions and stuff or yeah um force matching tasks um um and i made it this more to the story there um but but um but people have tried to link that to like um delusions about agency um and things like that so that's a just general point being that um that the the way that uh kind of computational mechanism is um you know looks like it happens in in one say sensory domain um can still operate very differently in the other sensory domain even in the same um in the same disorder right in the context of the same like um so so these things don't don't necessarily they shouldn't be expected to to generalize across the cross domains um is one one point and the other the other point was um you know in relation to you know things like hypnosis which um i don't know a lot about um as part of a textbook chapter um i was asked to write a paragraph or two on hypnosis so i kind of looked into that literature a little bit um and it's actually pretty messy they're very different there are multiple different theories about hypnosis work um and i mean one of them you know can relate to this idea that um that um you know if if the hypnotist repeatedly like gives suggestions that are set up such that um they will be confirmed right so it's hold out your the hypnotist says hold out your arms and then he says um your arms are going to start to feel heavy and then they'll start to fall right obviously

anybody who holds out their arms they'll eventually start to feel heavy and they're they'll you know heavier and they'll start to drop just as a result of your muscles your shoulder muscles getting tired right so if you do a number of those things and the kind of uh things the hypnotist the predictions made by the hypnotist are confirmed then you know it could be that you're just coming to assign a really high reliability to whatever the hypnotist says right so so in that case um if if the hypnotist just comes to be assigned really high precision then you can just prior expectation adjust prior expectations in a way that just kind of dominate um or can end up having a much stronger influence on what you do and that's that's kind of one you know one idea but again there's there's several other theories of hypnosis that just have to do with kind of like conforming to um kind of social roles and things like that that you're told to um that you're kind of told to do and i mean anyway so i wouldn't necessarily feel that confident um saying much about about how this relates to hypnosis other than that in a broad sense it can involve manipulating um sort of tricky ways of manipulating for our expectations in other in other people but again i mean even that is pretty speculative i mean so if i could just quickly throw something in another complicating factor so even within the visual system having a strong prior in one domain or doesn't generalize to others this is really great paper that just came that came out i think last year in cognition where they looked at representational momentum moony images and um goodness i think there was one other i think was illusory contours and they had one other top-down expectation kind of paradigm and there were large individual differences between all of them and there were also large individual differences in kind of the effect of prior expectation across those different tasks um so i think there's just kind of a lot of complicating fact like moving parts and a lot of this stuff that we have to deal with when we talk about these things okay we're going to return to the stack and go to dean and then a question from the chat and then adam so i'll go back and be ambiguous um so when i was looking at your diagrams and one of the things that really i wanted to ask this question but we ran out of time last week um do you think we can say that if a person says i see this or i do not see that see this the or yes or no um can we can we apply an idea that there's a certain amount of acceptance and a certain amount of letting go and i i kind of bundle that under we reconcile or is it something completely different i mean how do you how do you look at this now when a person's actually looking at something and going okay i do not see the box is it just is it that simple it's a simple yes no or do you think there's something more built into it how to there is a whole cottage in this cottage industry i feel like college industry might even be an understatement on how to collect verbal reports um and the proper ways of doing it so

there's like a perceptual awareness scale where you in some sense have a description of the content so you have i didn't see anything i maybe saw something but there was no specific content like you have a vague feeling you saw something but it was a bit blurry and then you had like a full lived experience that's one way you can have people place bets that's another way you can have people do confidence ratings lately there's been some incredibly clever work basically looking at so just give a shout out to people who whose work i really admire like hakun lao and um megan peters have a really great paper in e-life where they basically look show that people seem to have optimal in the sense of like bayes optimal introspective access to their own performance so basically when they are they will people never behave correctly um without kind of knowing that they behaved correctly as it were if you do these like very tightly controlled experimental situations um so yeah i think maybe i was a bit flippant before when i said you have to take reports as foundational i stand by that statement but um there is like years and years and years of like careful psychophysics they've gone into being like yeah these things are reliable um and basically it comes down to i wouldn't believe some if if someone said yeah i see something but they were getting like 51 like 50 on a forced choice task about the orientation of gabor patch but they were claiming they would say i wouldn't believe them generally speaking you would expect if you see something your performance should also go up that is not to say however that you can't have um before increased performance without awareness there is some convincing evidence showing that those two things dissociate and that's kind of what we were trying to show here right right yeah okay so hopefully that answers the question yeah it does nice yeah i noticed on this extended taxonomy the bottom right it's like consistent prior high attention strong signal it's the perfect storm and then it's 100 seen 100 correct and then again kind of taking it loosely mapping to our experiences with other attention phenomena like if you're sure about seeing in it of course the way that you tuned prior or inconsistent prior uh these are variables that are in the model so people can play around this isn't like a specific yes no but just examples of how rich the phase space is in terms of what the model can do and what it can capture so here's the question from the chat and then it will go to adam so the question from the chat is any thought about the possibility of using this model to investigate lucid dreaming related phenomena like the failure of implicit metacognition while dreaming thank you so how about dreaming or lucid dreaming so that stuff is super interesting i mean i don't think this model captures that at all except in like the very in like a computational sense i don't know what task would be modeling specifically but like we can if we're being a bit metaphorical about it um so

presumably i don't know anything about lucid dreaming i presume it's a case where basically you are dreamings you have kind of you're not just asleep you have kind of a experience that goes along with it and also you are aware of the fact that you are dreaming um that's really interesting i think it would have to come back to the i don't know how we one would capture metacognition in this framework i think that's one thing to say i don't know what it would be maybe ryan you have some thoughts on this but it would relate to basically how how one thinks about metacognition in these deep temporal models yeah i mean i mean i mean metacognitive itself comes in a bunch of different flavors right i mean so there's kind of you know beliefs about uh how good your memory is for example right it can be kind of metacognition it's you know it's like beliefs about your beliefs or patterns in your attention or there's a bunch of bunch of different types of metacognition right it's just sort of beliefs or operations over representations of your own cognitive processes so i mean the the tricky part is that you kind of have to have a higher level that's treating what you chose to do uh you know in terms of a cognitive voluntary cognitive process that observes that and then infers something about it um and um you know the way that that ends up influencing the the the cognitive the cognitive operation level below right the way that that higher metacognitive level ends up um setting priors on the lower level um i'm not sure the right way to think that i mean you'd have to look at literature on the way metacognition affects you know performance and cognitive tasks which i'm not um i'm definitely not one thing to say just very briefly would just be in the discussion section of our paper we discuss alternative models um i think probably the model that's closest to ours in the literature would be steve fleming's higher or the state space model um and i think there he explicitly discusses ignition related phenomena and kind of metacognitive inferences about the presence or absence of stimuli so i would say if you're interested in that kind of stuff and how to think about it in a basic framework look there it's a really great paper yeah i mean another another one i mean that comes to mind i know is that um there's a paper by um where the first author is claus stefan um from 2016 where he talks about um the way you can apply um active inference models to model um homeostatic and allostatic control processes so basically like anticipating for example that your temperature is going to drop and therefore increasing temperature body temperature a little in advance so that when it subsequently drops you stay in a um within a kind of survivable homeostatic range um i'm just using temperature here as an arbitrary example of any kind of variable internal bodily variable by like glucose levels or or hormone levels or anything like that um but um they do talk about this level a metacognitive level that you can add above the

allostatic regulation level that more or less tracks the efficacy or the kind of success of the allostatic level um and how if the allostatic level repeatedly fails to keep homeostatic variable or keep bodily variables within homeostatic ranges then the higher level can basically infer that allostasis will fail um and they show how that can possibly lead to certain sorts of symptoms of depression and you have low beliefs about the efficacy of your own allostatic uh processes um so i mean that's a that's a paper that i'd recommend looking at for one example of the way that you might apply one type of metacognition on top of the basement i also just remember the paper came out yesterday i shared i i sorry i forgot about this i just shared it i shared it on twitter yesterday if you want to look um it came out of mike allen's group um i've gotten the first author's name i'm really sorry about that um but uh they explicitly discuss like predictive processing and active inference theories of consciousness in relation to interoception and metacognition and they have a whole discussion of this type of thing so if you're interested maybe just like check my twitter page or at mike allen's twitter page sounds good um we'll go to adam and then anyone else who raises their hands hi um perhaps tying a few of these uh things together um i was wondering if with respect to hallucinations and psychosis and forms of delusions if um some of it you could potentially model it in terms of um not having access to mental acts so which might have like associated efforts copies that could potentially like be adjusting precision and or also having uh different components of the associated mental act which you may or may not attribute to you like some so aspects of interoceptive inference that could lead to senses of like willing or ownership but then by not having access this potentially where the access itself uh of the action generation could um basically adjust what you're how you're attributing um what sense you're making of the associated perceptual events so it's like um how did i say this so it's like if you're uh talking to yourself in thought or something like like thought is in her speech like vygotsky uh let's talk about like well if you don't have access to like the generation part and for some reason i don't know how much it can be separated but then like the voice might just like appear to you in a way where you don't have a sense of ownership if there's not like the right type of like maybe intercept of coupling to the genera generation process or just like a lack of metacognition to contextualize anyways yeah i really like this idea i mean one thing and one thing to keep keep in mind here is that you know the the the sort of thing that you're when people have sort of different experiences of or were there a different contents and experience i mean basically that's where you're talking right different contents that can be inexperienced those those are just in a model like this going to amount to different sorts of state spaces that you can infer

posteriors over right so you know you can have beliefs how much ownership you have over your body or you can have beliefs about something going on in your body interreceptively and all those things are just gonna be you're gonna have some sort of afferent signal and you're gonna have some prior expectations obviously and then you're going to infer some posterior over whatever the contents are of the thing in question right so so anything along the lines of what is the what is the content of one thing versus another is just going to amount to a particular level in a model that infers posteriors over particular right so none of that is going to have to do directly with our model other than the fact that given posteriors over any content um you know if our model were applicable to that you could just treat those contents as the lower level right so you'd have inference over some content and then our model would say something about um the processes that make that content conscious or not right or consciously accessible or not so so the only thing that our model might be able to say is if you swap out the you know square versus squares versus lines in our model right uh whatever the beliefs are at the lower level you swap that out with representations of some other content whatever whatever it is you're interested in you know something about the body or something about um what uh you know what a pattern of what sequence of the speech right that you're hearing or anything like that right what what word you just heard um you know swap that out as the content of the lower level and our our model might be able to say something about the processes that determine whether you whether those contents become accessible or not but beyond that our model doesn't have much just that yeah i mean if i could say something on that as well i i really like the idea i think just along similar lines to ryan because it just casts like decisions about ownership is just having access to some content or another and so for me like the essence of any type of conscious access or awareness process is basically whether it's integrated into this temporally deep representation and in doing so available to all of these sub-processes our global workspace theory um and so i think this kind of relating this to kind of the metacognitive point i think this is actually a spot where maybe this is a difference between owl theory versus an active inference theory that gave an explicit role for um so for me it's just the fact that you can be aware that it is you're aware of it because it's integrated into this representation right um whereas maybe higher order person b no you have to also infer that you are seeing it or that i am seeing it or something like that um and so i i um i think there's actually like really interesting empirical questions that you're what i'm going to say um but i i hope that we can kind of i don't know maybe it would be nice to see a discussion of kind of active inference theories of visual of metacognition versus active inference theories of

like visual awareness in general yeah i know if you i know you've written a lot about kind of different theories of consciousness adam so i know if you have any thoughts about that i mean i know that um you know there's this preprint um by um the first the first authorism is smith lars um well i can't remember uh uh his first first name but but uh but his last name is smith um that um does propose a a type of model of medic dog mission cognition there's in theirs attention works a little differently than in ours but essentially you do have a second level that controls a precision parameter on the um on the first level and then there's a third level in there that infers things uh about um control of uh the control of the temperature parameter at the first level um if i'm remembering right um and they use it as a way of the way they talk about it is a way to think of um like meditation and mindfulness and things like and things like that um which i'm not you know i don't remember enough detail to to say a lot about you know my my specific feelings on that but that is a an example that i'm aware of again it's just a preprint at this point but um of another attempt to try to model um something like metacognition cool interesting questions adam did you want to say something else yep go ahead um yeah so um i basically want to throw your model at everything but um in terms of uh metacognition and what i've worked on in the past mostly i've been focusing on like trying to cross-reference various theories to see the points of like overlap and non-overlap mostly considering like accounts of dynamic modularity that might apply to a global neural workspace architecture or like integrated information theory where you can think of their complexes of integration those modules as themselves being like do an iit handling of those and seeing like where's the overlap basically trying to give semantic content to iit via like gnwt is like an architecture for bayesian model selection and the whole thing kind of being adjudicated by energy principle and mostly an active inference not nearly the technical depth that you have i'm only beginning to move into things like metacognition and uh like higher order forms of and the ideas are like really uh speculative like so like one idea is as you're like um let's say um imagining like enacting something you might start out from like generating like experience from an egocentric perspective but then this enactment would auto-associatively also have third-person allocentric representations like from like the ventral visual stream or something like this like you've seen people or yourself doing similar things and so you might then like be able to get access to in the mind's eye this like third person point of view on you with some unfolding like some sort of like moving back and forth between a first person like i guess and the models i'm working with like phenomenal consciousnesses all and and accessible consciousness there's a sense in which like in a rudroff um uh projected geometric modeling it's always from this first

person essential point of view but then the idea is you then are looking at an objectified you and this would be part of metacognitive awareness and contextualizing you in terms of like you doing actions and then this third-person little homunculus doing actions that you're seeing in your mind's eye and moving back and forth that's like the basic space i'm moving into i don't know how much of that's going to work or or how you would model that well i mean like one thing just kind of kind of kind of um i mean i mean again just to just uh i mean when you think about this kind of informal terms right then then if you're in some sense um if you are the content you're describing as a belief that you are doing something right um so that's just the content in the state space right so same kind of thing right swap out our lower level with the state space that is beliefs about what i am doing right then it's just the same thing right you have a representation about what you are doing and in our model you would either um you would either gain access to the representation of that content or not via its integration with this deeper temporal level um granted beliefs about what i am doing you know or not is already right pretty pretty temporally deep um but were those sorts of contents to apply to our models and that's what it would say it would say that um those contents need to be integrated with this deeper temporal uh model that is able to form posteriors over something that incorporates that um and uh so again i mean at the end of the day really it really just comes down to this distinction about the content being represented and whether or not the posteriors associated with that content get integrated with the with the higher with the higher level that does the sorts of things that the higher level described here can do so so it's just a question about whether something like that is the content that you want to be uh using in a goal director we are recording um or or or whether just uh just representations of the of the lower level kind of visual content right and that being integrated with this level is is what's is what's involved so it's me again this just has to do with different levels of representation and what the contents are that you're becoming conscious of um so oh and by the way i just just uh um i just mentioned i looked up the paper i was mentioning um so it's it's called towards the formal neurophenomenology of metacognition and i'm the first author is lars sanbed smith um just to um do justice to uh uh remembering who the first author's full name was in the name of the paper i was talking about so um sorry i think i think you started saying something at the same time chris yeah christopher's then adam i was going to change the topic very slightly but i i was wondering so how you think about phenomenology or phenomenal consciousness because to me it comes down to the nature of the representation that you're gaining access to so to talk about like you talk about the projective consciousness model which is really cool there this

perspectival nature of consciousness comes from the structure of the generative model and the fact that they're using this projective geometry and that's built into the structure of the generative i'm just curious about how you think about this and how that kind of maps on to iit because at least as i understand iit like phenomenal consciousness is kind of a i i've never really been clear how the content of consciousness fits into iot like it's all uh um anyway yeah i'm just be interested to hear your thoughts how it all fits together with like global workspace like um yeah i would say there's some kind of giant begged questions in iot um i think though it's not nothing that they start from um axioms of what they think should be part of phenomenology and then work their way to this sort of way of handling systems in terms of i think it's a decent prior for like what we should look for in like physical substrates i wouldn't like completely like throw it off but they then would say if you have these axioms that characterize experience some intrinsic existence uh composition information uh integration exclusion if you have these uh different properties then it is sufficient to bring about phenomenality because they started from phenomenality oh no i don't think that follows at all um that being said um i think it does potentially like give us some prizes and trying to think about physical and computational substrates of consciousness so that would be like the cross-referencing and then the other part would be like how do you think so if you're thinking global neural workspace as like potentially like a trading off of modularity as being part of the physical implementation well then the question's like so when do you have like larger big modules functioning as like dynamic cores or workspaces and when do you have like more fragmented like local processing so like in theory you could get like um bayesian model selection with like discrete updating with like a bunch of small beta complexes close to the modalities or it could be like this big sprawling alpha complex that's like going that's multimodal and all these different ways but then like you're moving back and forth between these like degrees of synchrony seems like iit could potentially be useful there for describing that so that was one of the ideas of that in a relation um in terms of like kind of bringing it to rudroth like for me like the object i would want that would be like potentially a minimal condition for phenomenality would be some sort of um joint uh distribution over your body pose and visual spatial awareness um as like the minimal thing for me at least for human-like consciousness and a series of um basically roughly at alpha frequencies a series of estimates of that and and and then and thinking of this as iterative bayesian model selection via a global workspace architecture but not necessarily one where there's like access that's like a more sophisticated kind of consciousness in terms of you wouldn't necessarily i would call it like it's

pretty darn global you have something like spanning all the posterior cortex but it's not the kind of work space you're dealing with where you're actually having knowledge and access thanks adam so if anyone else wants to they can raise their hand otherwise i think there was a very good request just you did just ask them so let's mix it up but um there was a very good request actually if uh ryan or christopher you could just look at this part of the figure which i know is a structure we walk through in the model stream as well and just kind of map some of these bigger ideas that we've been talking to some of the letters just in a way where people can now look at this outline say okay the experiment i'm kind of on the page we talked about a bunch of visual and interreceptive kinds of experiments just where did it map in this paper so that people will always be able to go back to this paper look at the code and then kind of map some of these bigger metaphors to the experiment that you did here um do you want to take this ryan or john i don't mean either either way it doesn't doesn't matter to me go for it um so how how detailed you'd like me to go do you like me to explain what the little square what the d's and a's and b's and s's are and all that stuff let's go for all the letters on the right side and maybe even some on the left all right so this is a partially observable markov decision process and it's a hierarchical one so at the let's just focus on essentially let's start at the bottom o's are observations and s's are hidden or latent states and what the a matrix does is it provides a mapping between those two things so circles are random variables and the squares are essentially fac uh functions in this case probability distributions that map between those two things so the a little arrow going down just says that that's essentially a with that little a superimposed on arrow is basically saying this that's expressing a conditional probability distribution saying that o observations depends upon the hidden state now as we start to evolve through time we need transitions between discrete hidden latent states in the world and those are what's described by b so this is something like what is the probability that at time one uh what's the probability of each hidden state at the next time step and it will describe kind of s at t conditioned on s at t minus one now then what's special about the hierarchical models is that you have another layer on top of that um where state hidden states at the first level are now being treated as observations by the second level and so the posterior probability at the first level is basically acts as an observation um for the second level a matrix um and then kind of at the second level there should be a d at the first level too but that's fine um basically what this prior what this second level hidden state does is it provides a prior over that first level and then at the first level you can then have basically we just have one time step at the first level um but you can have any amount of time steps as you want so if you imagine the metaphor

that's often used is like uh the minute versus the hour clock on the hand they're evolving at different time scales um so you could in this diagram we show that the second level evolves at two time steps for every one time step at the first level at the second level sorry um at the second level there are again transitions between these latent states and that's also where we have policy selections that's why we put it up there so policy selection is essentially if you imagine a whole series of hidden markov models this is my favorite way of describing policy selection so these hidden markov models are basically just a partial observable markov decision process without the decision aspect if you imagine a whole series of basically these graphs and you are deciding what graph kind of is going to be my future you are trying to find the graph with the highest model evidence that is what computing doing policy selection does so you are choosing the transitions that will take you to basically the graphs of the highest model evidence uh i know that brian you want to kind of clean up some of that explanation we're at the shoulders let's go to the last few variables yeah i just like chris said right close it's just your distribution over policies um g is the um expected free energy um so that's the essentially the the function that decides right which um what the value is you can think about it about each each policy and that is with respect to c which is um your preference distribution which is which is essentially the the thing that specifies which observations you want and which observations you don't um now um somehow i see that an e has been placed with an arrow down to g um and that's not correct um e should be e should be going you should be going straight to π um if you're going to include e um i don't think we did anything with e in our model so i don't think e needs to be there um but um but e if it's used is just a kind of separate prior policies that can encode something like habits um so it competes with the expected free energy g over policies um to infer what to when you're inferring what the posterior river policies is um so uh so yeah i mean otherwise i feel like chris uh i feel like chris described it well i mean you know it'd be a little it's always a little easier if you have you know the if you're in control of the pointers you can actually you know point to uh i think you also asked us to relate this to an experiment to her paradigm so if we maybe briefly click down to figure two what you've labeled as figure 2 there so we can just look at that so the idea here is that there was a forwards mask at the beginning of the paradigm this was just a series of lines and there were these kind of stimuli on the outside at the second time step in the task the original like simulated task they on some trials the square kind of was self-organized the sum of the lines kind of self-organized into a square so i either in our discrete state space model a square was present or it wasn't and in the third time step it was just replaced by more lines and

then afterwards we asked the agent to construct a report of that so basically at you can think of the first level of the model as being all the features of the stimulus and also the location of attention so this might be something like saliency maps in posterior parietal cortex which are directing where attention is being pointed to you might have and then you might have the features of the stimulus which are represented in various places at the second level we now have our this is basically our fronto-parietal cortices this is tracking the evolution of states at the first level so it's tracking that at the first time step there was lines presented at the second time step a square presented maybe it maybe it was maybe it was present maybe it wasn't and at the third time step there was more lines and you can imagine in like a real task all of that unfolds in a space of about a second yeah so you can think about it i mean yeah the second level is really the the content in question that it's inferring is the sequence right it's not whether the square it's whether it was lines square lines or just lines lines lines um and so it's another and so if you're thinking about where experience lives obviously we don't experience a sequence right i think and i think ryan i've been chatting about this a lot lately because it relates to work we're currently doing but i think what's going on is that what we experience are essentially like the updates to our second level beliefs and that's really crucial so the reason why it's crucial is because we know empirically that kind of what could be we might describe as brain processes they're at the first level of the model these can be either conscious or they or not um and contents that are conscious when you are conscious of something it has all of these really important functional consequences so here it might be kind of being able to see it for report maintain its report but also if i see something i can then voluntarily maintain it in memory for as long as i want as long as i don't get distracted like there are limits on like human psychology um but that that's kind of a really important function that's enabled by being conscious of something right and that allows us to do things like construct reports of our experience or even i i think by report i just mean something very general it could be in this model we kind of have the toy thing of being able to like put a sentence together but it could also be more more empirically real um in terms of empirical paradigm will just be like hitting a button or doing a confidence tar doing a confidence interval or something like cool thank you for that interesting answer we'll have um stephen and then adam and then anyone else who wants to raise and also any last questions from a live chat as we kind of slowly land the plane yeah um i was just going to ask in relation to those lines and the way that appears like you said so this this is actually like a graphic representation but when someone looks at it there's probably some special device that you've got that represents this is it like that the lines

rotate um to be and at some point and it just happens they rotate such that they line up and then they un-rotate or and just one other question is why didn't you use negative space like you like say they line up to make you know like a negative space square i just wondered whether that relates to the choice of how it pops in and out of perceptual punch sorry

phenomenological phenomenological consciousness or access um so basically this is just a very in our model we have discrete hidden states which are categorical distributions which are lines or squares and external states which are red and so this is kind of just a graphical representation of a way to think about that but it has really nothing to do with how it works mathematically um then in terms of actually the empirical task like this is this is a task that we basically like borrowed from michael pitts um who's a really done he's a cognitive neuroscientist on a really important work in the area of visual consciousness and as for why they chose to kind of have these i guess the i think from memory there are videos attached to this paper from memory i think the lines just kind of they jitter around and then occasionally a subset of them will all line up into a square and then go back to like jittering um as to why they did that versus other methods like i don't know really they definitely are subtle things you can do to a stimulus which have major impacts on whether conscious access to it or not but the phenomenon of the inattentional blindness is super general like these are you can have people so the really famous example is that everyone gets shown in like undergrad psych is the ever up there's you have your tracking whether a group people playing basketball and you ask how many how many times does someone like a white shirt get past a ball or something like and your attention is distracted so kind of consumed with that that uh a bear or a gorilla can like walk into the middle of the screen being the dead center of your fovea wave at you and then walk away and then about 50 of people don't see it at all um and that's really well replicated across a um or they don't report seeing and i think i should say i think there's some there's some debate about whether this is a memory phenomenon or any of that there's lots of subtleties there but um so hopefully that answers the question yep oh also one advantage and one rationale for these pure modeling papers that we're discussing today is the space of the possible human experiments or even any other kind of real world experiments is very limited it's very limited what you can actually have a real set of humans do and so it's great to have tools that help us explore some of the patterns that we're looking for and understand how variability and do statistical uh calculations like how many participants might we need of a given uh variability range if we want to capture such and such an effect without distorting it this way that's critical information and if you don't have the then the experimental design

phase is totally blind and it's a very um shot in the dark and so this helps inform structures for even thinking how to design human experiments for as we're talking about here with visual awareness but it could be for other kinds of experiments so adam and then anyone else who wants to make a comment see you later alex bye adam uh hi um i'd be curious in uh knowing more of your thoughts on the potential i guess uh richness or lack thereof of experience so like at one point like could consciousness be phenomenal or more experience as a series of snapshots that are like static sequential but we don't know it like a flip book or is it more like a continuous stream if you're getting this discreet updating are these updates over like like the agent and like motion like if you take like a camera sometimes in your phone there's like a little brief like forward head like look ahead and back but like what would be the nature of these uh discrete updates yeah i mean i know the like i'm trying to remember i know um and i remember reading um dana hans uh book um a few years ago and there is there is some work i think showing you know there are kind of there's like a minimal time for like essentially updates to the um to the of consciousness to have some kind of discrete character to them i don't um i don't remember what psychological refractory period yeah do you know what that do you remember the actual like milliseconds i can't remember like 50 milliseconds or something like that yeah i don't i don't remember a fan but there is there is something you know like that maybe might be you know like similar to like a refresh rate or something like that um but i yeah i don't remember too much about that off hand um but but i mean one thing you know one thing to say is that um you know because you know there are these sorts of discrete updates in our model but i mean it's also good to remember that the the higher level representation is specifically about this um sequence right um which is integrated in a sense that the hypothesis is about a whole sequence that doesn't necessarily need to be thought about as having sort of discrete chunks to it right it's just this tingle uh just a single hypothesis about what the what the whole sequence was and so you might think about that right as having a more kind of continuous character to it despite the fact that it's it is updated in some ultimately discrete fashion you know and kind of these really fast bins um you can think about it that way um but but and in terms of richness i mean to me i mean i just think of i i just tend to think of rich as meaning there are more precise features right there's more features that are more precise um at the at the lower level right so instead of just you know a lines or a square right at our you know in our example um there could be a ton of different lower level representations about color and shape and size and you know all these sorts of things um that are represented in some sort of joint way right and become accessible um and and so i mean that's one way to

think about richness is just what what are all the things being represented that become access that become accessible um together um but i don't know they used to ask a question actually there about the continuous modeling is it possible that active inference could have a continuous mode um i know that it would get rid of a lot of the discrete benefits and the sequential message passing or maybe other heuristics but could there be a continuous format continuous timer space yeah there are continuous state space models and there are also mixed models um you know mixed models are kind of especially nice and probably a lot more realistic here um you know because visual visual input a lot of the things that are represented by the visual system are continuous right like uh motion for example is continuous brightness um you know all these all these sorts of things um and so those can be those can be um perceived and represented in a continuous scale at the lower level but still get passed up to dispute representations at the higher level um we didn't do that here for simplicity but you could sorry um i was just to clarify i was in scaling active inference we did talk about the continuous state space i was just wondering if it's possible to have a continuous time active inference model rather than a discrete time model of t_{123} just just bringing it up okay but then christopher and then blue so yeah three things so just quickly to answer your question daniel um so hidden markov model is just like a super general thing it's basically just when you have a markov chain and but you have but each state of the markov chain links to some outcome those outcomes can then both the outcomes and the latent states can be continuous when you have a discrete latent state continuous outcome you end up with a gaussian mixed with like basically what ends up being a mixture of gaussian's model um and when you have if you have continuous states you end up with uh i think you basically end up with a kalman filter and you can stack those on top of each other and do all the same things it's yeah for all the reasons you say things get really complicated when you start to move to continuous time and um but i completely agree with ryan i actually i think there are computational there are functional reasons to think that decision making is discrete and so at the decision making the level of decision making computationally speaking we should use discrete state space models i also think that at a certain level we need continuous state spaces obviously we represent continuous quantities and i think thomas parr we've chatted about this before in the model stream but thomas par has a really nice paper on kind of the discrete continuous interface and that's something i'm actually really interested in i'm not sure i don't know to what extent we should take these things as idealizations or as um general kind of neurophysiological predictions i think that's a really i'm super interested in that in other words but i don't know another thing to

say though is that it's not as though only um only action you know only action selection has to do with um discrete state spaces there there are possibly a lot of higher level um representational things that are also industry so for instance like concepts right like i can i can have a bunch of continuously represented lower level visual features but i can use those to infer something discrete as well like for instance that those features correspond to my concept of a dog right or my concept of a banana or you know whatever right because we do have these discrete categories that we map on patterns of continuous features too um and i mean that's a that's a kind of whole another discussion i think we talked about abruptly last time right gaining conscious access to the visual features versus gaining conscious access to the fact that it is a dog right so those are different levels of representation that can be attended to and you know may be able to kind of uh independently or semi-independently be accessible um separately um so but that's just i just want to point out that not everything not everything is going to be continuous in in perception when when sort of passed probably and that's where yeah in fact as you can see that's where a lot of category theory discussion comes into play let's do blue then christopher for final thoughts so i i know like we operate under the assumption that uh the visual like we perceive things continuously in visual space but i mean really the input is like 60 hertz right so i mean theoretically like this thing this object is consistently a dog but if it's like flashing back and forth to a cat at sub-perceptible levels like we wouldn't know right so so i mean i think anything can be modeled in discrete um chunks if you break the chunks small enough pieces yeah yeah i've sometimes seen people make be a bit smarmy about this when um someone says like oh discreet state space is unrealistic and the comeback is always like mate like you can always discretize things up to some arbitrary level and then discrete state spaces work fine um i know i actually think that's a bit of a dodge to be honest because the way we're using these discrete state space models is in like an ultra discrete kind of way right so i think that if we had like if we had a hidden markov model where we had like six a 60 hertz equivalent or something that would just be ridiculous like you should just work in continuous state spaces at that point like the the the computational and conceptual advantages of working in a continuous discrete state space are gone it's so much it would be so difficult it's a really nice question though and it's like if you do make it a finer and finer granularity you keep some of the really big benefits of splitting and sometimes it's actually easier to even do like a protein folding they'll do a time step of the tiniest tiniest amount and do millions of time steps are tiny because it's still easier to fit that ultra rich discrete time model with actual time steps that can be clustered on different computers rather than re-write

the whole base to do continuous modeling so i agree it's not like continuous is simply better and it really relates deeply to how we think about the continuum and the infinitesimal and so it's really an interesting area for active so steven and then any other closing comments yeah i suppose if we also were to go back into the physics of it you know with active inference as a way of like if vision is feeling the surfaces out there rather than it being an input signal even if we have these brain waves some of that might be an artifact of cognitive science which has got like the input process output at some level it's it's extracting like alex constant talks extracting you know information from quantum noise noise you know random fluctuations and all this sort of stuff so you know there's it could be it could be that discrete state space is kind of you know sort of also because there is quite choppy what's coming in and at some point inferences have to be made that makes it more like a signal if that makes sense now i'm just i'm just putting that out there just to that as another layer at the lower levels of the retina and stuff like that good any last comments otherwise this was super interesting i guess a closing question for the authors would just be when when's the next uh episode in this paper saga or what's the next what's the next thing you're excited about here um we have we have another paper that's kind of i don't know maybe like three-quarters done or something like that that's kind of the next yeah so they're done it's an expanded model that does a lot more with um so it allows for a no report paradigm and um selected selective attention and working memory maintenance are our explicit policies um in the updated one there's a number of other advantages but you know it's coming soon yeah i think it's a matter of uh me riding it really sorry wow too real but thanks everyone for joining this is really fun for 18 overall and we look forward to probably seeing you again for 19 and beyond thanks to all the participants you can fill out the survey for feedback in your events calendar invitation otherwise we'll uh be talking through other channels so thanks everyone see you later thanks